

Smart Compaction: Predicting Compaction Utility from Lakehouse Table Metadata

Jannic Alexander Cutura^{1,2} Subash Prakash³

¹European Central Bank ²DSTI School of Engineering ³IBM

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Outline

- 1 Motivation
- 2 Methodology
- 3 Results
- 4 Cross-Schema Generalisation
- 5 Discussion
- 6 Conclusion & Future Work

The Small File Problem

Lakehouse table formats (Iceberg, Delta Lake, Hudi):

- Append-only, log-structured writes
- Each commit adds new immutable Parquet files
- Snapshot isolation + time travel

Consequence: progressive file accumulation

- Query planning scans every file's metadata
- S3 LIST/GET calls are priced per request
- Excessive file-open overhead degrades scans

At Scale

Catalogs with thousands of tables accumulate **millions** of undersized files.

AutoComp (LinkedIn): 35,000+ Iceberg tables.

Compaction

Rewrites many small files → fewer target-sized files.

But compaction **costs compute**.

When is it worth it?

The Gap

- All open-source scheduling is **threshold-driven**:
 - Iceberg: purely imperative (no built-in scheduler)
 - Delta Lake: triggers on small-file counts
 - Hudi: commit-count / elapsed-time rules
- AutoComp (LinkedIn) uses heuristics + multi-objective Pareto ranking; its authors name ML prediction as future work
- Proprietary systems (Databricks Predictive Optimization, Snowflake auto-clustering, AWS S3 Tables) are **not open or reproducible**
- **No published study** examines *which* metadata features determine compaction utility

Our Goal

Predict compaction utility from table metadata alone, using ML, and identify which features matter.

Problem Formulation

File count reduction ratio:

$$r_{\text{fc}} = 1 - \frac{n_{\text{after}}}{n_{\text{before}}}$$

Two tasks:

- 1 **Classification:** Does the table need compaction? (did bin-packing rewrite any file, $n_{\text{rewritten}} > 0$)
- 2 **Regression:** Predict the continuous $r_{\text{fc}} \in [0, 1]$ for cost-aware scheduling

Key constraint: features extracted only from Iceberg metadata (manifests, snapshots, partition stats).
No data files read.

Parameter grid: 7 axes, 2,376 combinations

- Row counts: 10^4 to 10^6
- File-size targets: 8 KB to 128 MB
- Partitions: 1 to 100
- Writers: 1 or 5
- Batches: 1 to 20
- Skew: uniform vs. Zipfian
- Payload: 5 or 50 UUID columns

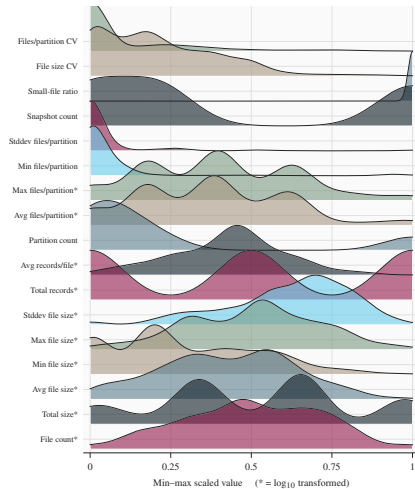
Heterogeneous writes:

- Batch 1: 50% of rows (initial load)
- Even batches: $8\times$ smaller file target
- Breaks the “tautology trap”

Compaction: Iceberg bin-packing

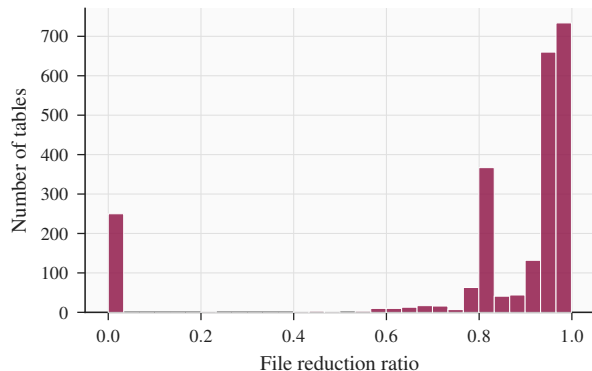
- Target: 128 MB
- Min: 96 MB (75%)
- Max: 230 MB (180%)

17 Metadata Features



All 17 features, min-max scaled to $[0, 1]$. Asterisked features are $\log_{10}$ -transformed. Extracted from manifests alone — no data file is opened.

What the Simulation Produced



Distribution of r_{fc} across all 2,376 tables.

- **2,376** tables generated
- **2,121** (89.3%) had at least one file rewritten
- r_{fc} : mean 0.818, median 0.950, $\sigma = 0.294$

Bimodal, not continuous

Either compaction does essentially nothing, or it does a lot. Only **11 of 2,376** tables land in $(0, 0.5]$.

So the binary label and the continuous target nearly coincide — which is the first hint that *whether* is the easy question and *how much* is the interesting one.

Classification: Binary Decision

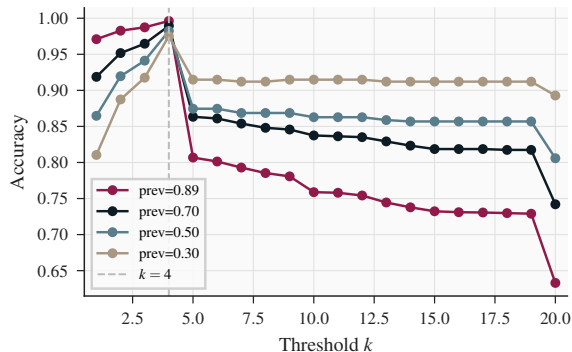
Classification (held-out test, $n = 476$)

Model	Acc.	F1
XGBoost	1.000	1.000
Logistic Reg.	0.815	0.885
Threshold ($k=4$)	1.000	1.000

The whole classifier

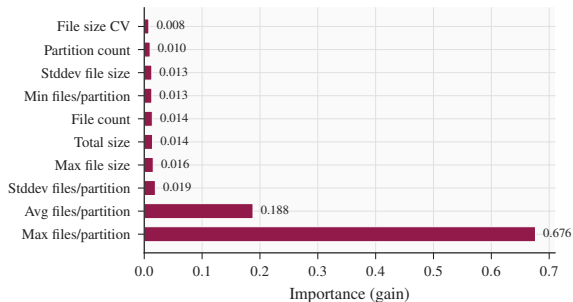
`max_files_per_partition > 4`

A single threshold matches XGBoost exactly. No learned model required.

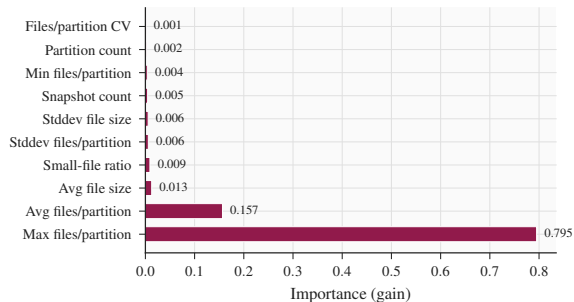


Accuracy peaks at $k = 4$ and collapses at $k = 5$ — a hard boundary, not a fitted one. Holds at every prevalence level.

Where the Signal Lives



Classifier — **86.4%** of gain



Regressor — **95.2%** of gain

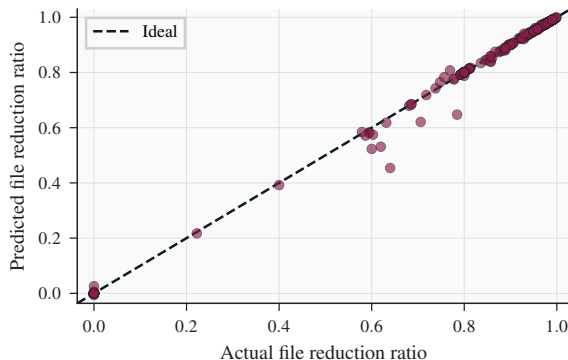
Two partition-level counts dominate both tasks. Every other feature contributes ≤ 0.019 gain.

Regression: Continuous Prediction

Regression (held-out test)

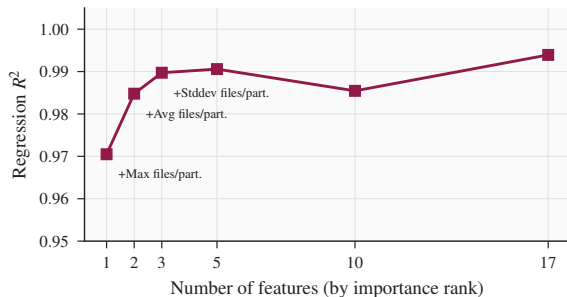
Model	RMSE	R^2
XGBoost	0.013	0.998
OLS	0.236	0.361

- OLS cannot capture the nonlinear saturation curve
- Two features account for **95.2%** of regressor gain
- Spearman $\rho = 0.965$: monotonic but not linear



Predicted vs. actual r_{fc} on the held-out test set.

Is the Result Fragile?



One feature alone reaches $R^2 = 0.970$; two reach 0.985. Past five, returns diminish.

Four robustness checks:

- **Ablation:** the other 15 features add only $\Delta R^2 = 0.023$ (95% CI [0.004, 0.044])
- **Leave-one-out:** every single removal still keeps $R^2 \geq 0.988$
- **30 bootstrap splits:** $R^2 = 0.999$, CI [0.996, 1.000]
- **Prevalence 89% $\rightarrow$ 30%:** regression $R^2 > 0.995$ throughout

Holds across model families too — random forest 0.998, decision tree 0.997. Any tree-based model suffices; only the linear baseline fails.

TPC-H Validation (No Retraining)

96 Iceberg tables from 6 TPC-H schemas (lineitem, orders, customer, part, partsupp, supplier)

- Realistic data types (strings, dates, decimals)
- 7–16 columns per schema
- Scale factor ~ 10 (468.8 M rows, 15 GB)

Cross-schema results

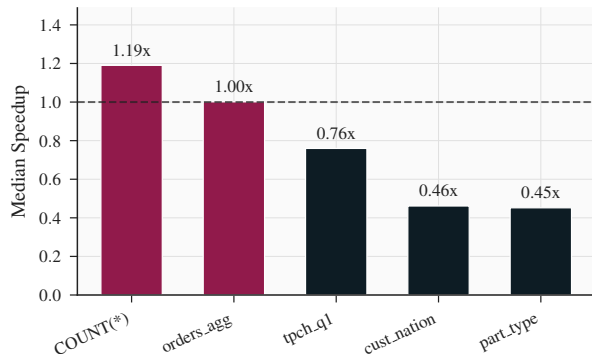
Task	Acc./RMSE	F1/ R^2
Clf (XGBoost)	0.979	0.987
Clf (Threshold $k=4$)	0.979	0.987
Clf ($sfr > 0$)	0.833	0.908
Clf (Majority class)	0.823	0.903
Reg (XGBoost)	0.058	0.976

Key findings:

- Transfers with **no retraining** — no TPC-H table was seen during training
- XGBoost and the $k=4$ threshold tie at 97.9%, missing the same two borderline tables (3 files at ~ 50 MB)
- The learned model earns its keep on the *continuous* target ($R^2 = 0.976$), not the binary decision

Weakest schemas are `customer` and `partsupp` ($R^2 \approx 0.87$), which produce tables sitting closest to the compaction threshold.

Does Compaction Actually Make Queries Faster?



52 TPC-H tables, Iceberg time-travel (VERSION AS OF) to query both snapshots. Median of 3 timed runs.

Not universally.

- COUNT(*): **1.19**× faster — metadata overhead dominates
- Filtered aggregation: neutral to 0.76×
- Full-scan aggregation: **0.45**× — 2× *slower*

Why

Spark assigns tasks per file. Compacting 300 files into 1 drops parallelism from 300 tasks to 1, offsetting the metadata saving for queries that must read everything.

Overall median across 104 measurements: **0.97**×.

Key Insights

Tautology Trap

Our first grid produced only small files (max ~ 0.4 MB vs. the 96 MB threshold), so *every* table compacted completely: $r_{fc} = 1 - 1/N$. XGBoost scored $R^2 = 0.99$ by recovering that identity ($\rho = 0.996$). Any benchmark using only small synthetic files will produce misleadingly perfect metrics.

Feature Dominance

Partition-level file counts (`max_files_per_partition`, `avg_files_per_partition`) dominate both tasks.

- Binary decision: fully solvable by a single threshold
- Continuous prediction: requires nonlinear modelling

Practical Implication

AutoComp's compaction traits already include these features. They are sufficient for the binary decision — but **cost-aware scheduling** needs the continuous prediction. A workable rule: compact when predicted $r_{fc} > 0.3$, which drops the lowest-utility jobs while keeping 99.8% of the beneficial ones.

Threats to Validity

Setup:

- Synthetic UUID payloads compress differently from real columnar data
- Partition structure generated, not domain-derived; query patterns not modelled
- Spark-local execution $\neq$ distributed clusters with S3
- 89.3% prevalence may inflate accuracy — mitigated by majority-class baseline, downsampling, bootstrap CIs

Scope — what we did not test:

- **Delete files / MoR:** we use Copy-on-Write only. Merging delete files is a different dynamic our features do not capture
- **Sort and z-order:** bin-packing only. Their utility depends on query workload, which metadata cannot see
- **Temporal evolution:** static snapshots; a table that is fine now may need compaction after the next write
- **One compaction target:** $k=4$ is tied to the 128 MB/96 MB configuration

Conclusion

- 1 **Binary compaction decision** is trivially solvable by a single partition-level threshold
- 2 **Continuous reduction ratio** requires nonlinear modelling (XGBoost $R^2 = 0.998$ vs. OLS $R^2 = 0.361$)
- 3 **Transfers across schemas** without retraining: 97.9% on TPC-H (threshold and XGBoost alike), regression $R^2 = 0.976$
- 4 Compaction utility is determined by **structural metadata**, not data content
- 5 Downstream benefit is **workload-dependent**: metadata-heavy queries speed up, full-scan aggregations slow down

Future work:

- Extend to Delta Lake and Hudi
- Add query-workload features and cloud-cost metrics
- Multi-table scheduling via contextual bandits
- Sort and z-order compaction strategies
- Validation on production Iceberg catalogs

All code and data publicly available.

Thank You

Questions?

Jannic Alexander Cutura		Subash Prakash
<code>jannic.cutura@dsti.institute</code>		<code>subash.prakash@ibm.com</code>

Backup: Per-Schema TPC-H Breakdown

Schema	n	Clf Acc.	Reg R^2
lineitem	16	1.000	0.997
orders	16	1.000	1.000
part	16	1.000	1.000
supplier	16	1.000	1.000
customer	16	0.938	0.874
partsupp	16	0.938	0.867

`customer` and `partsupp` each contribute one misclassified table. Both sit near the 96 MB compaction threshold, where a small file-size shift flips the outcome.

Backup: Query Latency by Type

Query	n	Median	Mean	ρ	p
count_star	52	1.19×	1.31×	0.50	<0.001
orders_agg	12	1.00×	1.63×	0.89	<0.001
tpch_q1	13	0.76×	0.82×	0.77	0.002
cust_nation	15	0.46×	0.56×	0.45	0.095
part_type	12	0.45×	0.47×	0.38	0.226

52 TPC-H tables, median of 3 timed runs after 1 warmup. Speedup $> 1\times$ means faster after compaction. ρ is the Spearman correlation with the file-reduction ratio.